Unit 10 - Communication Technologies
Assignment 6 - Upgrade a LAN Tasks 5&6&7

Introduction

Task description - *You are a junior consultant for T5 Alliance. Your new client, Competition Bikes Ltd, have decided that they would like to have wireless capability in their new network. You must adjust your design to include the following new requirements;*

- *Wireless coverage must extend to the whole site.*
- *The wireless network must be as secure as possible.*

Task 1 & Task 2

(U10P6) - Describe the most commonly used transmission media / (U10M2) Build on instruction 5 by describing a network scenario where each of the media types would be best utilised, or what the most common use of each media type is.

Task description - *Description of a common situation where the following transmission types are used; / Description of a common situation where the following transmission types are used;*

Note: (On this assignment, I have combined the first and second task due to the fact they are consistent with each other, fitting my writing style. Meaning that I can write about the two tasks together while also (hopefully) achieving both the criterion instructions under the same subheading)

Introduction

With data communication technologies, transmission methods are a variety of different methods and techniques which can be used to transfer data/signals from one location to another. Due to these differences, each have their own methods, advantages and disadvantages.

Different transmission methods vary substantially, this is because of the number of different methods, and these include both bound (wired) and unbounded (wireless) methods of transmission.

Coaxal

The coaxial communication method is a bounded transmission media, meaning that it is a physical connection. Consisting of copper and a non-conductive material used to separate the inner and outer conductor. The outer conductor is there to conduct the interference made from the central conductor.

With coaxial, the data travels through the center conduction within the cable, where the interference is caught on the outer copper mesh. So, in terms of transmission, the transmission of data is done by the use of two concentric conductors which are separated by an insulator.

The coaxial transmission media is a bounded media, meaning that it is a physical connection. Consisting of copper and a non-conductive material used to separate the inner and outer conductor.

The outer conductor is there to conduct the interference made from the central conductor, which gives it the advantage over other methods, because others may not be used because they cannot guarantee no interference to the signal from other sources.

Coaxial cable may be very simplistic and cheap in design, but it does have a variety of uses when dealing with data transmission.

For instance, it is used as a transmission line for radio frequency signals, used in businesses and other organisations as the ethernet cable and has many other uses which fit into a local area network.

Most commonly however, Coaxial cables are used for radio and TV broadcasting, due to their ability to transmit audio and display signals. However, it is becoming less common because there are other, more modern forms of transmission replacing coaxial.

Overall, coaxial cables are used for their cheap and wide availability. Essentially, coaxial cables are also used because of their diverse application (they are compatible with most electronics), low cost and high reliability, and they are used extensively worldwide.

Furthermore, Coaxial wiring is very inexpensive compared to other methods, and also doesn’t allow interference to have an impact on the quality of video and image due to the design of the wire, including an inner conductor for data and outer copper mesh to absorb interference.

However, with Coaxial transmission, the wires are generally large in size and they can sometimes be hard to install and remove because they screw onto the electronic device, but this does mean that they are more securely installed, making it less likely to come out loose by accident.

Optical fibre

Optical fibre cable is a bounded transmission method, consisting of a fragile, transparent fibre made out of glass, which has the width of less than that of a single human hair, coated with multiple layers of plastic.

Fiber optic cable uses electrical signals to transmit data, by using light with a variety of different colours, to change the wavelength. Meaning that in a fiber optic cable, light only moves in one direction. Therefore meaning that the wire isn't very flexible, and that it needs to be set up with care, also due to it being fragile.

Fibre optic cable could sometimes contain two strands of cable, where each stand is responsible for one direction of communication. A laser at one device sends pulse of light through this cable to other device, then these pulses translated into binary at the receiving end.

Due to the specific purpose that optical fibre was made for, it is only typically used for telecommunication or supply high speed internet connection. Meaning that it doesn't have much versatility and only restricted with the use of ISPs and networks, such as LANs and WANs.

The reason that it is only used for specific networks is because of the unconventional ports optical fibre uses. Since it uses light to transmit data, it used ports widely different and less common to that of others, such as twisted pair cables.

Overall, optical fibre transmission has a very high bandwidth available, especially when compared to other methods of transmission. This is beneficial because this allows more data to be transferred in a short time period.

Optical fibre also has very good security, this is because data being transferred by the use of optical fibre cannot be intercepted by outside parties. However, one big disadvantage of optical fibre is that the installation is expensive.

The use for optical fibre is limited only to networks, but it does carry out its purpose very well, transmitting data at impressive speeds when compared to other cables, but comes with the cost of being fragile and more expensive.

Unshielded twisted pair (UTP)

UTP is a common type of cable which consists of two wires, which are twisted around each other multiple times, this is to cancel out electromagnetic interference from other external sources. Due to their cheap costs but efficient data transfer, UTP cables are the most common cable used in computer networking.

The reason that UTP is so common is due to the fact it is so cheap but durable and able to transmit data at a decent speed. For instance, if a user is going for the cheapest method possible, then a UTP would be recommended due to the cheap costs and decent performance.

In a network, UTP cables can be used to connect different hardware together, such as hubs, switches and workstations. Since it is such a simple wire, it does not mean it has a simple use,

since it could be used on a wide variety of hardware, whether in a large network, or just connection a PC to a router.

However, UTP cables do have one intended purpose, which is to connect different hardware together, so they can communicate together. Since it is such a broad intended purpose, there is a wide variety of machines that can use UTP, especially hardware that is found within a network.

Overall, A UTP is an effective choice because it is cheap to install, however does not offer the same bandwidth and security as other methods. Basically, the reason that UTP's are used in certain situations because the cable is lightweight, thin and flexible, as well as versatile, reliable and inexpensive.

A disadvantage of UTP is that the bandwidth isn't as high as other methods, such as optical fibre. Also the level of security isn't as secure as that offered by optical fibre, this is because this cheap to install, meaning that the cable does not support very high bandwidth or offer security due to the lack of insulation from the signal bleeding.

Since it is such a simple and cheap cable, it can be used within a network for a variety of connections. Whether it is connecting a server to a HUB or a workstation to a switch, the UTP is a very reliable cable to use due to it's durability, from the twisted wires creating a much stronger hold, and flexibility.

Which can't be found in other cables such as optical fibre, because it is too fragile and has a bend limit due to its method of transmission. However, UTP doesn't have the durability and safety found within STP, which I will talk about next.

Shielded twisted pair (STP)

Which brings me onto shielded twisted pair, or STP. Similar to UTP, STP is a common type of wire in which consists of two copper wires which are twisted around each other multiple times, just like UTP.

Unlike UTP however, is that STP are shielded by a form of insulation, such as an additional copper braid jacket or foil wrapping, which helps protect the signals on the cables from external interference.

Basically, STP is just like the UTP, but consists of a form of shielding over the wire, making it less likely to create a fire hazard and reduces the levels of electromagnetic interference, improving the performance of the data transmitted through the wire.

STP is more expensive than UTP but has the benefit of being able to support higher transmission rates over longer distances. STP is similar to UTP but with each pair covered by an additional copper braid jacket or foil wrapping.

While STP's are more fragile than UTP as the shield must be kept intact in order for them to work properly, the best use for STP cables are in industrial settings with high amounts of electromagnetic interference, such as a factory with large electronic equipment, where they can be properly installed and maintained.

The use of STPs are similar to that of a UTP, which is used to connect hardware found within a network together, giving them the ability to communicate. However, since UTP has a shield over the wiring, which can be waterproof and fireproof, it reduces the fire hazards that UTP could previously bring.

Overall, The advantages of using this type of transmission is that it carries data at a very high speed, this therefore means that this type of wiring is suitable to be used in Ethernet networks, especially those which operate at a high speed.

A STP offers better security than an UTP however still falls behind fibre optic when it comes to bandwidth and security. An STP also offers better security, due to its improved insulation from outside interference, which therefore gives it an advantage over a UTP.

However, it is more expensive to install because the wires are more expensive due to the extra production needed to shield the cable, but it doesn't mean it isn't worth the extra pay.

The reason being that this shielding helps protect the signals on the cables from external interference and reduces the risk of fires within a network, which is possible to other forms of cable such as UTP when put in the wrong areas.

Infrared

Infrared is defined as an invisible energy which is contiguous to the red end of any visible spectrum and contains electromagnetic radiation in the wavelengths which are between 800 nm 1mm in length.

Computer infrared network adapters both transmit and receive data through ports on the rear or side of a device. Infrared adapters were installed in many laptops and handheld personal devices. Infrared connections were be created through the same method as other local area network connections.

Infrared networks were designed to support direct two-computer connections only, which can be created temporarily if the need for infrared is apparent within a network. However, extensions to infrared technology supported more than two computers and semi-permanent networks.

IR wireless can be used for both short and medium range communications and control within a network.
What I mean by this is that some systems operate in line-of-sight mode. this means that there must be a visually unobstructed straight line through space between the transmitter, being the source, and receiver, which is the destination.

Other systems operate in diffuse mode, also called scatter mode. This type of system can function when the source and destination are not directly visible to each other. An example is a television remote-control box. For instance, the box does not have to be pointed directly at the set, although the box must be in the same room as the set, or just outside the room with the door open.

Overall, infra-red is cheaper than other similar technologies, therefore making it desirable. However, infrared only works over a short distance, such as a few metres, and only works in the line of sight of the object.

Making the range of infrared a significant disadvantage and should only be used as a data transfer if the devices are in the same room. So, infared is only used within a network if the need for one is apparent, and seems to be more advantageous than other forms of transmission.

Radio

Radio is a form of transmission of electromagnetic signals through the air or free space, an example of one of these signals would be use of sound waves. Essentially, a radio transmits data with or without line of sight, meaning that transmission can take place anywhere, as long as it is between the sending antenna and receiving antenna.

In a network, typically a type of WAN, radio transmission is chosen in particular situations because it doesn't need cables to send data therefore making it cheaper and more efficient than other forms of transmission.

Radio transmissions are used because they are able to communicate on long distances without having to use wires. An example of a use that radio has in networking, is by a Radio Access Network, or a RAN.

To elaborate, a radio access network is a technology that connects individual devices to other parts of a network through radio connections. It is a significant part of modern internet access, with 3G and 4G network connections for mobile phones being examples of radio access networks.

Which in order to receive any transmitted data the user will need to adjust the frequency on the radio in order to pick up the appropriate channel, at that moment the radio frequency would capture the appropriate transmissions from the wireless signals being streamed across the location.

Which is why sometimes it's possible to pick up different radio channels in certain locations because the radio waves don't reach and the location may have a radio channel on the same frequency.

Overall, It is a simple connection because the fact is the radio station is the only device transmitting the radio waves and the radio is receiving the waves based on the frequency selected on the radio.

The technology that is commonly located within networks in the radio frequency band is Bluetooth, infrared, Wifi, 3G and 4G connections, due to these connections not requiring a wire.

However, radio is risky to use because it can become unclear when the signal drops, this be affected by the whether etc. However, an advantage radio is that in enables the user to communicate with others without the use of wires.

Microwave

Microwaves are a form of electromagnetic radiation in which the length of the waves differ in size, which is used in technology that uses high frequency beams of radio waves to provide high speed wireless connections that can send and receive voice, video, and data information.

The reason being because microwave transmissions can easily pass through the atmosphere, making it usable for Satellite TV, smartphones and other mobile devices with the ability to connect to this network freely, without the need for physical cables and it that it can be connected from a decent distance.

Also, microwaves can have a use within networks. For instance, Metropolitan Area Network protocols, such as those based on the standards such as IEEE 802.16, is designed to operate between the frequencies of 2 and 11 GHz.

This gives the ability for mobile phones and other forms of portable devices use microwaves to connect and transmit data. Which is done by a small antenna, giving the smartphone and over devices with this technology the ability to connect to wireless connection which uses microwave transmission.

However, microwave transmission also requires line of sight in order to work properly. In order to allow two way communications two frequencies are used, but this does not mean that there

has to be two antennas, because the frequencies can be dealt with by one antenna at both ends.

Microwaves are effective for their use, this is because they can work in a large area, this is ideal for items such as mobile phones because this ensures that they can be used in a vast amount of places, also microwaves require a small antenna in order for them to connect, this also makes them ideal for mobile phones since this allows them to stay a small size.

Microwaves are the most effective transmission method used for mobile phones because other alternatives such as radio waves because they can be affected by the whether etc.

Overall, An advantage of the use of microwaves is that no cables are used in the transmission, multiple channels are also available and data is transmitted at a high bandwidth. Meaning that the user can access the network freely without the need of cables, reducing physical limitations.

However, the range of microwaves are still limited, like any other form of transmission. So the user has to be in the desired range to access the microwave frequencies, at least to a sufficient enough standard to get enough signal strength for data transfer.

A disadvantage of the use of microwaves is that if the line of sight in obscured, such as a building being locating in the way the signal will be disputed. Also, microwave towers are expensive to build, making the initial cost of a microwave significantly large,

Satellite

Satellites can be used for communication, the definition of a satellite is an artificial device which orbits the earth for the purpose of communication. However, a satellite can be used within a network to connect two different LANs over a large distance, creating a WAN. This is called Satellite Internet.

To elaborate, Satellite Internet is the ability to transmit and receive data from a relatively small satellite dish connected to a router within a network and communicate with an orbiting satellite found above most of Earth’s atmosphere.

Basically, a satellite operates the exact same way as a microwave tower, but the difference being that a satellite is orbiting around the planet, usually through the equator. Meaning that it is covering a wider signal than a microwave tower would, due to the signal expanding down to earth, instead of a vertical signal in between the atmosphere and earth.

The orbiting satellite transmits and receives its information to a location on Earth, called the Network Operations Center, or NOC. The Network Operations Center is usually connected to the Internet , otherwise a private network. So all communication made from a satellite dish to the orbiting satellite will flow through the NOC before it reaches the Internet, meaning that the signal can be redirected with ease.

Basically, what I mean by this is that since satellites are in orbit, the signals can be sent instantaneously into space, and then redirected directly to the network's destination, using a satellite dish, or even another satellite, creating a chain reaction.

A basic example of this transmission media would be a digital TV signal. As the satellite acts as the middleman or more like a channel when used, the satellite is streaming radio waves that contain different channels on them, back down on earth.

Which when a user switches a channel the satellite streams the appropriate wave based on the channel chosen and the benefit is that if you notice satellites have a fast response time because the channel changes without any delay. This same logic applies to the use of internet signal throughout a network full of workstations, since it can send VoIP, video and over forms of binary data found on PCs.

The Speed for satellite transmissions goes up to 400Kbps, but it might be able to go further thanks to technology improving, and plus a satellite is known for dealing with multiple systems at one time. Meaning that having a small amount of satellites isn't as significant of a problem, otherwise satellites would be obsolete.

Overall, communication via the use of satellites allows for a high bandwidth connection, fairly cheap to use, but not for the initial costs of a satellite, and also cover a large area due to their position.

However, satellite transmission does require an unobstructed line of sight, due to the fact that satellite signals must travel in straight lines but do not have the limitations of ground based wireless transmission.

Since the line of site will be between the orbiting satellite and a station on Earth, there isn't much higher of a chance of interference due to the fact the signal on earth would be too spread out to interfere with completely.

With this wide area coverage, satellites are useful for situations where someone is broadcasting live content around the world. However, one massive drawback to the use of satellites would be that they have a huge initial costs, due to the production for the original device and launching it into orbit.

Another disadvantage would be that they can suffer from large amounts of interference if the unlikely chance of debris covering the satellite dish, which therefore affects their quality or a lack of overall signal.

Task 3

(U10D2) - Compare the effectiveness of any two of the transmission media you described in instruction 5

Task description - *Comparison of TWO media types in terms of;*

- *Performance/Speed*
- *Security*
- *Cost*

Two media types: Optical fibre & Satellite

Performance/Speed

Optical fibre is well known for it's impressive speeds, so it isn't surprising that the speeds of optical fibre is much faster than the performance of the satellite. Especially when satellites have a much larger distance to cover than fiber, making it more possible for interference in part of the signal.

To elaborate, there are apparent problems with satellite which are much more likely to happen than the problems found in fiber optic.

For instance, satellite suffers from propagation delay, meaning that the signal would be lost for a temporary amount of time, reducing the overall performance of the network. Giving a significant impact on the performance on the signal, making it very frustrating to use for the user and increasing the chance of the signal crashing/not responding.

For optical fiber, no such delays exist, meaning that it is impossible for optical fiber to suffer from similar delays. Thus fiber optic is quicker and has higher data rate than Satellite in data communication.

However, optical fiber is much more fragile than a satellite signal, which means that the performance of fiber optic will come to a complete halt when the cable breaks, bleeding out the vital light used to deliver the data.

For satellite, The performance of satellite communication is very slow, due to the distance that the signal has to travel compared to the speed that the signal travels, to reach the destination point.

Adding latency to the connectivity, which is a very important aspect to consider on networking because even a few seconds delay is enough to cause some problems with internet connectivity.

Overall, optical fiber's performance is much more significant and consistent when compared to the performance of the satellite's transmission media. This is not only due to the speed comparison between the light used for optical fiber and microwaves on a satellite, but also the distance and interference that both satellite and fiber can come across.

Security

Optical fibre also has very good security when compared to other transmission media. This is due to the fact the signal, being light, doesn't radiate from the cabling. Meaning that it is extremely difficult, maybe even impossible, to extract data from the optical fibre cable.

However, if the signal was to bleed from the cable, it would render the cable useless because the light, containing the data, wouldn't be able to reach the destination. Causing the entire system to malfunction. Which is why it would be difficult to extract data from the cable, especially if it was to be undetectable.

The security found on satellite internet signals is very poor, especially when compared to optical fibre cables. The reason that security on satellites are so poor is due to the ability for external interceptor being able to access the wireless signal emitted from the satellite.

While extremely difficult, it is still possible with enough skill, meaning that the network that the satellite is connected to, would be at risk of being intercepted by an external offender. While there can be security implemented within the network that the satellite is connected to, the satellite itself has very poor security, especially when compared to fiber.

Fiber optic network also gives better security management, because all the hardware and other devices are situated in one central location, instead of having multiple devices scattered around the building, each requiring individual security management from the satellite signal.

Overall, the security on fiber and satellite is completely contrasted because of the excellent security of optical fiber, due to it's structure, and the poor security of satellite, because of the way it functions.

Cost

The cost for long range communication, satellite communication is very low, especially when compared to the significant cost for long range optical fibre communication, excluding the cost of repair which is also very high for optical fibre.

Fiber optics is wired network and requires a huge investment in setting up than satellite, depending upon the size of the network. This is due to the fact that purchasing a high bandwidth speed is more expensive than the standard speed, which is required for fiber optic otherwise it would be obsolete to have it installed instead of a standard copper wire.

Also, a network will require a lot more optical fiber cables than satellites connected to the network, meaning that more investment is needed to purchase more fiber optic cables, when compared to just accessing the signal from a satellite.

For example, minimum three or four satellites can cover the whole world. While fiber optics it's nearly impossible to carry this out, and if it were to happen, the cost and considerations are to be worked out would be at extreme levels, which is only possible with solution communications.

However, the initial costs for fiber optics is extremely low when compared to the Satellite's ludicrously high costs to produce and launch into orbit. This is due to the fact that optical fibre consist of a cable with a glass tube very small in diameter, while the satellite is a large object that required a lot of energy and precision in order to launch into orbital distance.

Overall, fiber optic is more expensive to operate than a satellite, because of repair, maintenance and costs for new cables for the network. While a satellite the business or user just needs to connect to the satellite's signal in order to get access from it, costing very little to compared to fiber.

However, initial costs are completely in fiber optic's favour, due to the fact that fiber optic is only a cable that needs to be connected. While a satellite needs to be launched into orbital heights in order for it to operate.

Conclusion

In conclusion, both optical fibre and satellite both have completely different uses, due to both the transmission media working completely different to each other. Which is why they both stand out so well from each other with performance, security and costs.

With performance, the fiber optic is significantly better to that of the satellites, because of the difference between the speed of both transmission methods. With fiber using light, and satellite using microwaves, light being the much faster method of transmission. Meaning that, even with the distance difference excluded, there will be a much larger latency gap for satellite compared to fiber.

Also, satellites have a much larger chance to suffer from delays in the signal, which isn't possible within a fiber cable. The reason it can't happen to a fiber cable is because the light has to stay inside the cable at all times, if it experiences some sort of delay or signal bleeding, it would just stop working in general.

With security, optical fiber is much better when compared to satellite also. Which is also due to the way both transmission media work. With fiber optic always having the signal kept within the

cable, with no signs of signal bleeding (otherwise it won't work). It would be impossible to intercept the signal of the fiber optic cable externally.

While satellites send signals down to earth in such a large radius, giving it the ability to be accessed a lot of places within it’s range limit, which is quite large. Increasing the chances of an unknown source intercepting the signal, and hacking into the network (if built with poor security) and even into the satellite itself.

With costs, the initial costs of satellites are much higher than fiber optic, due to the setting up of the satellite requiring a significant amount of resources, especially when compared to the setting up of fiber optic.

However, to use both signals, the satellite is actually cheaper because it doesn’t require the cost of faster internet speeds, repair costs and maintenance costs which the optical fiber cable requires in order to function properly.

Overall, optical fiber is much more desirable than satellite when considering these specific terms, due to how fiber is outstandingly better than the satellite due to the way they both function.

However, this does not make the satellite obsolete, because there are other attributes which the satellite can do better than fiber. Otherwise, if the satellite were obsolete, it wouldn’t be used at all in modern transmission media, and optical fibre would be used instead.